

Purpose

This guide/analysis was created to introduce the Fortinet based network architecture by providing core knowledge of the FortiGate firewall console and graphical user interface (GUI). This guide also gives insight into the default configurations of a FortiGate device which gives important experience in managing network security, which is a core skill for any networking administrator who has to deploy, secure, and maintain standard SOHO networks.

Background

Small Office/Home Office, aka SOHO, is a network type commonly used by individuals or small businesses with less than 10 employees. This network type commonly uses smaller scale network devices compared to their large enterprise counterparts. SOHO networks provide numerous advantages to smaller teams as they are easier to set up and are more affordable than full size network equipment. SOHO networks often only have a single router, and even can contain switches, wireless access points, and end devices such as computers and printers.

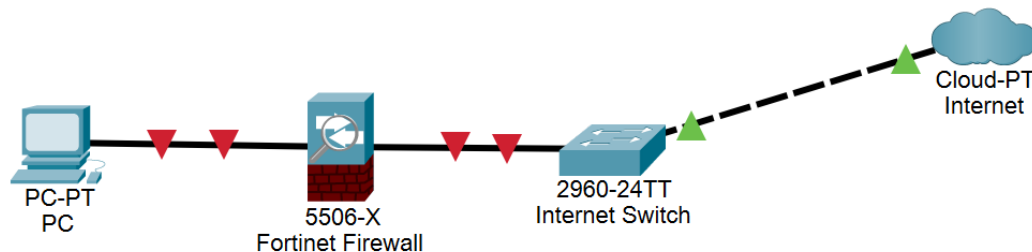
There are 4 types of networks: Personal Area Networks (PAN), Local Area Network (LAN), Metropolitan Area Network (MAN) and Wide Area Network (WAN). In this analysis, we will be focusing on Small Office/Home Office which falls under the LAN category. Small Office/Home Office (SOHO) networks connect devices for resource sharing and internet access with the main components being the switch, wireless access point and router. The switch connects wired devices like computers and printers to the local network. The router is essentially the device that lets the network access the internet, while managing traffic and providing security.

A firewall is also an essential part of the network and focuses on security features. These security features mostly consist of monitoring and controlling incoming and outgoing network traffic based on a set of security rules. It prevents unauthorized access to the network from the internet. Historically, the early networks involved connecting computers with physical cables to a central hub for resource sharing and communication. However, today's SOHO networks are essential for small businesses and home offices to access the internet, share certain digital resources like files and securely connect to a corporate network if needed.

To fill out these strict harsh security requirements without introducing not easy to manage architectural complexity, organizations frequently turn to specialized vendors like Fortinet. Founded in 2000 and headquartered in Sunnyvale, California, Fortinet has grown into a core company in the cybersecurity industry, largely due to its philosophy of unified threat management. While legacy networking setups often piece together mismatched hardware

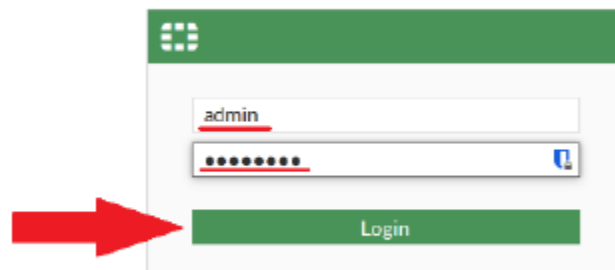
from various manufacturers, Fortinet centers its strategy on an interconnected ecosystem known as the Security Fabric. This approach ensures that its next generation firewalls work seamlessly along with secondary hardware components and cloud delivered security services. By having routing, switching, wireless management, and threat intelligence into a single ecosystem, Fortinet can directly compete with other industry leaders such as Cisco, Palo Alto Networks, and Check Point, scaling fluidly from SOHO environments up to massive enterprise infrastructures.

Topology

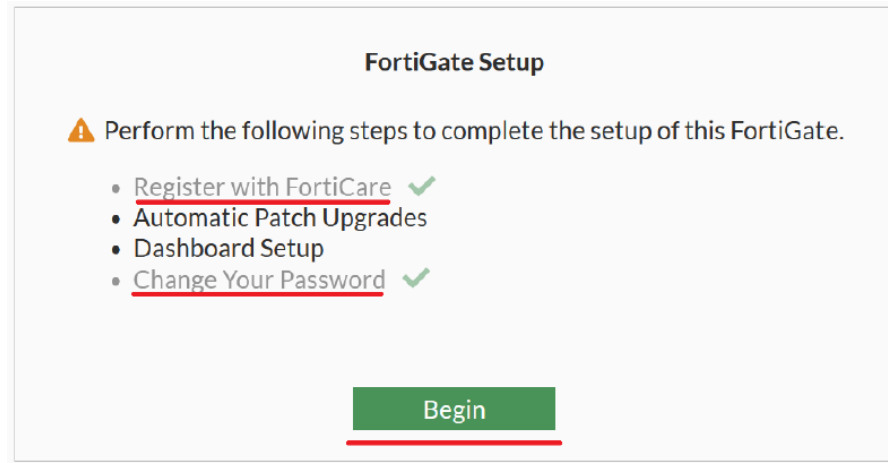


Lab Commands

Connect your PC to the firewall via ethernet. Go to <https://192.168.1.99> and log in with the username and password you set.



You should see the following setup screen. Note that the Change Your Password checklist item has already been completed, as we completed this earlier. Click Begin.



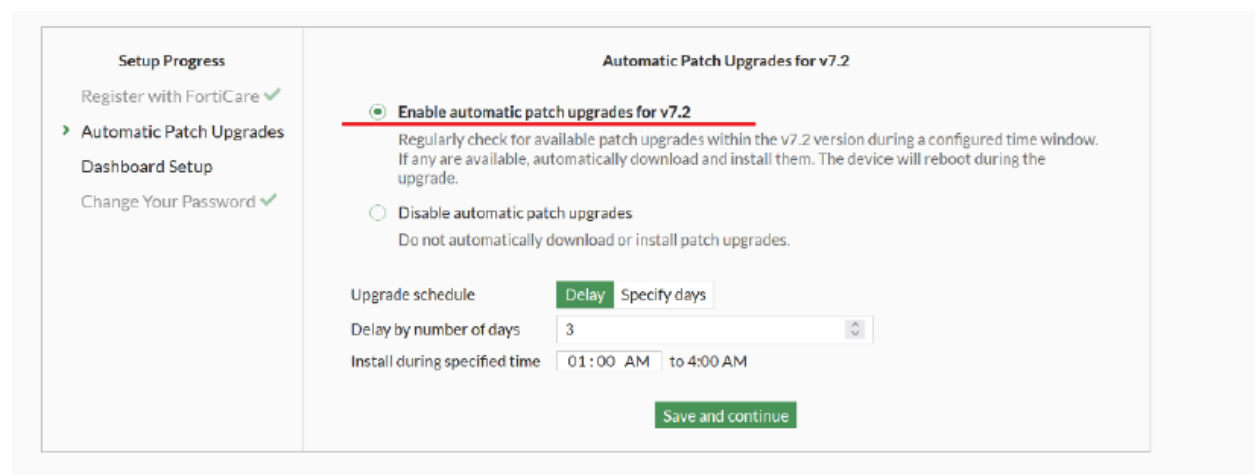
FortiGate Setup

⚠ Perform the following steps to complete the setup of this FortiGate.

- Register with FortiCare ✓
- Automatic Patch Upgrades
- Dashboard Setup
- Change Your Password ✓

Begin

Click Enable Automatic Patch Upgrades and leave the other settings as default.



Setup Progress

- Register with FortiCare ✓
- Automatic Patch Upgrades
- Dashboard Setup
- Change Your Password ✓

Automatic Patch Upgrades for v7.2

☒ **Enable automatic patch upgrades for v7.2**
Regularly check for available patch upgrades within the v7.2 version during a configured time window. If any are available, automatically download and install them. The device will reboot during the upgrade.

☐ Disable automatic patch upgrades
Do not automatically download or install patch upgrades.

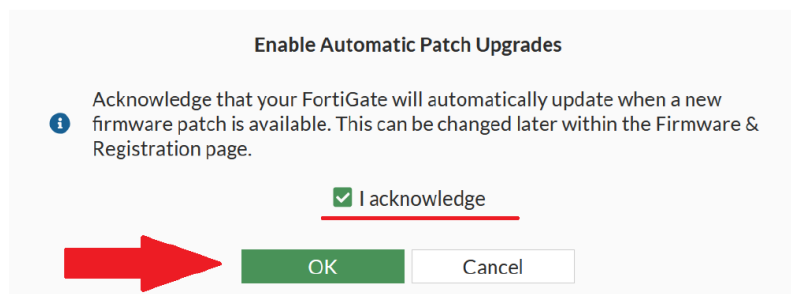
Upgrade schedule: **Delay** Specify days

Delay by number of days:

Install during specified time: to

Save and continue

Click I acknowledge and click OK.



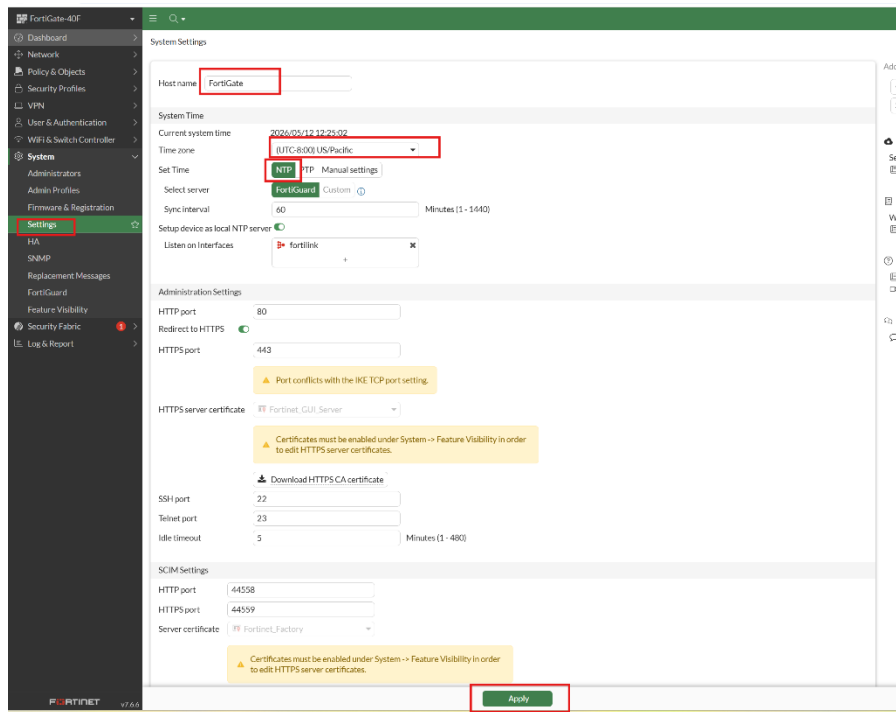
Enable Automatic Patch Upgrades

i Acknowledge that your FortiGate will automatically update when a new firmware patch is available. This can be changed later within the Firmware & Registration page.

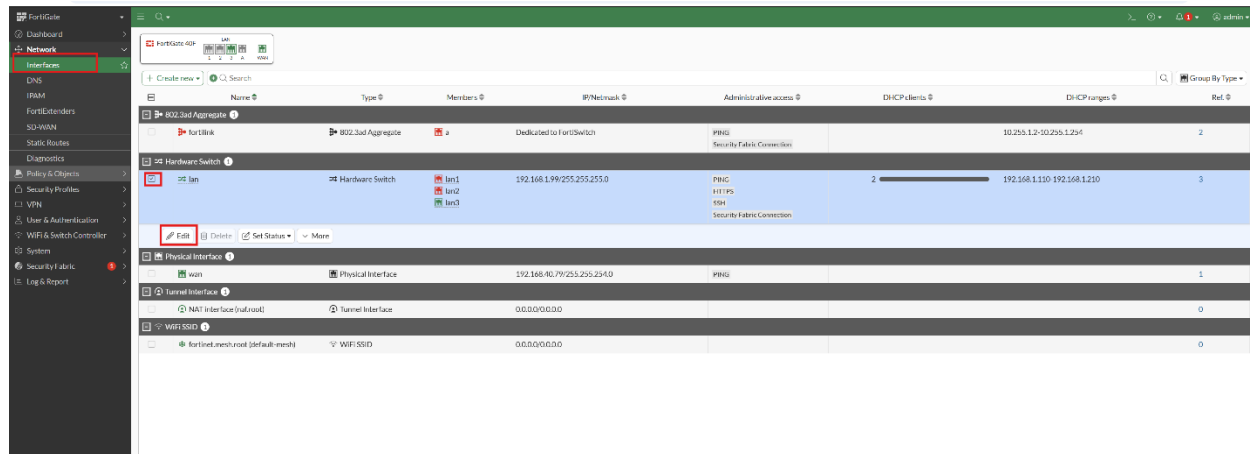
☒ **I acknowledge**

 **OK** Cancel

To change the hostname and configure the system time of the Fortinet firewall, head over to System, Settings and configure the Host Name and the respective time zones.



Next, head over to Network → Interfaces. Click on edit to configure the different parts of the network. First we will do the LAN interface.



Here you can configure the LAN configuration settings. For this lab, we will be configuring its Alias to Office LAN. You can also optionally change the IP address of the web-based firewall management interface to an IP address of your choice by configuring the IP/Netmask into your desired IP address. For this lab, we changed from the default IP address of 192.168.1.99/24 to 192.168.10.1/24. For DHCP, you can configure the address range for leasing. For demonstration, we have configured the address range of leasing from 192.168.10.10 to 192.168.10.200. Don't forget to set default gateway to the configured IP/Netmask. When you are finished, press OK to finalize your changes.

Edit Interface

×

Name

lan

Alias

Office LAN

Type

Hardware Switch

Interface members

lan3 lan1 lan2

Role

LAN

Address

Addressing mode

Manual PAM DHCP PPPoE

IP/Netmask

192.168.10.1/255.255.255.0

Create address object matching subnet

Name

lan

Destination

192.168.10.0/24

Secondary IP address

Administrative Access

IPv4

HTTPS

FMG-Access

FTM

Speed Test

SSH

RADIUS Accounting

SCIM

PING

SNMP

Security Fabric Connection

Receive LLDP

Use VDOM Setting

Transmit LLDP

Use VDOM Setting

DHCP Server

DHCP status

Enabled Disabled

Address range

192.168.10.10-192.168.10.200

Netmask

255.255.255.0

Default gateway

Same as Interface IP Specify 192.168.10.1

DNS server

Same as System DNS Same as Interface IP Specify

Lease time

604800 second(s)

FortiClient On-Net Status

Default Specify

Advanced

OK

Cancel

FortiGate

FortiGate

Active Administrator Sessions

HTTPS

Status

Up

MAC address

84:39:8f:f5:3a:29

Additional Information

API Preview

References

Edit in CLI

Online Guides

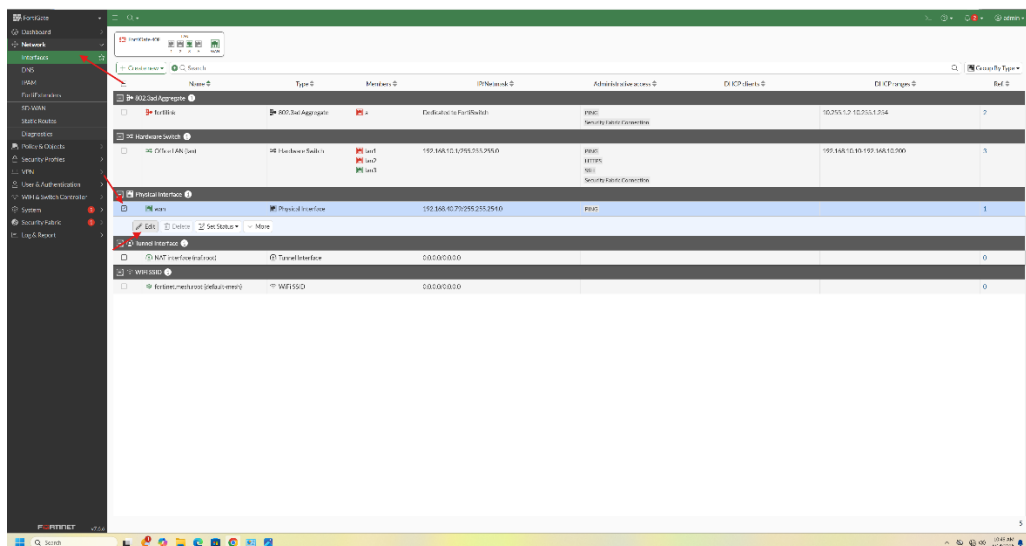
Relevant Documentation

Video Tutorials

Fortinet Community

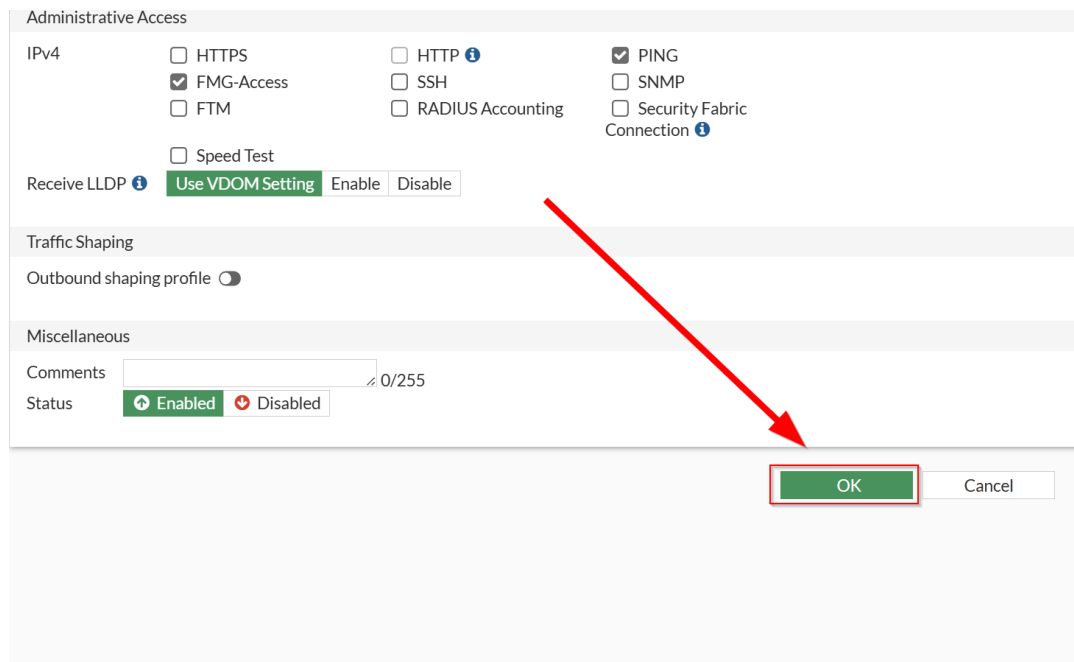
Join the Discussion

Similarly, by selecting the WAN and clicking on edit, we can also configure its Alias and interface members.



Name	 wan
Alias	Internet
Type	 Physical Interface
Role 	WAN
Estimated bandwidth 	0 kbps Upstream
	0 kbps Downstream

Press ok to save

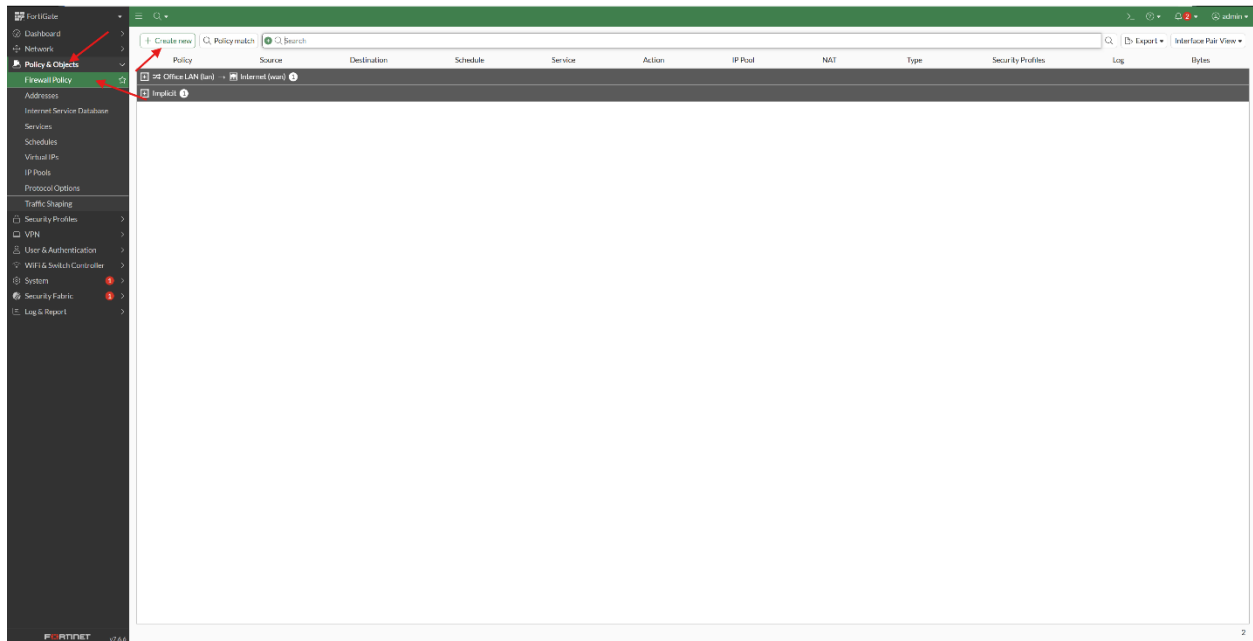


The screenshot shows a configuration window with the following sections:

- Administrative Access**
 - IPv4**
 - ☐ HTTPS
 - ☒ FMG-Access
 - ☐ FTM
 - ☐ HTTP ⓘ
 - ☐ SSH
 - ☐ RADIUS Accounting
 - ☒ PING
 - ☐ SNMP
 - ☐ Security Fabric Connection ⓘ
 - ☐ Speed Test
 - Receive LLDP ⓘ: Use VDOM Setting Enable Disable
- Traffic Shaping**
 - Outbound shaping profile: ☐
- Miscellaneous**
 - Comments: 0/255
 - Status: Enabled Disabled

A red arrow points from the 'Use VDOM Setting' button to the 'OK' button at the bottom right of the window.

To configure your LAN-to-WAN firewall policy, head over to Policy & Objects, Firewall Policy and press Create New.



Here, configure these configurations. This will allow LAN to WAN policies that direct traffic from all sources to all destinations, with NAT configurations. When you are done, press OK.

Create New Policy

Name: LAN to WAN

Schedule: always

Action: ACCEPT

Incoming interface: Office LAN (lan)

Outgoing interface: Internet (wan)

Source: all

User/group:

Destination: all

Service: ALL

Firewall/Network Options

NAT: ☒ Use Outgoing Interface Address

IP pool configuration:

Source port translation: Always When port conflicts Never

Protocol options: PROXY default

Security Profiles

AntiVirus: default

Web filter: default

DNS filter:

Application control: default

IPS:

SSL inspection: no-inspection

Logging Options

Log allowed traffic: ☒ Security events All sessions

Comments:

0/1023

Enable this policy: ☒

OK Cancel

When finished, your screen should look like this.

Get Connected to your default browser and join it to your toolbar

Firewall Policy

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	Type
LAN to WAN	all	all	always	ALL	ACCEPT	Enabled	no-inspection	UTM	6,821 MB	Standard
	all	all	always	ALL	ACCEPT	Enabled	default	UTM	0 B	Standard
							default			
							no-inspection			

You're done with your SOHO

Problems

During the lab, I found out that the Fortinet AP cannot be plugged directly into the FortiGate without a separate power source, since the FortiGate does not provide power over ethernet (PoE). We fixed this issue by placing a standard 48V PoE injector in between the firewall and the access point.

One other problem that I ran into happened after we manually set our ethernet IPs and the default gateway to the IP address of the Fortinet. After I set up the SOHO configuration, it didn't work. I looked at `ipconfig /all` and realized that I wasn't getting any DNS addresses. This was because I was supposed to be using DHCP automatic addressing instead. After I switched our settings back to DHCP, everything worked perfectly.

Conclusion

In conclusion, this guide/analysis gave me a strong understanding of the basic Fortinet ecosystem, the FortiOS GUI interface, and how to configure various WPA2 security policies. Overall, the setup was straightforward, and I am very confident that I could replicate this exact configuration in a real life SOHO network.